

BUILD OPS — Refactor PRD (Execution Edition)

Construction ERP for Actuate Builders Inc. (ABI) · built by Th/rd Code Solutions Inc.

Version 1.3 · 12 August 2026 · Supersedes v1.0–v1.2

Reference toolchain: OpenAI Codex CLI. Tool-agnostic in substance.

Companion files:

In the repo	Purpose
AGENTS.md (root)	Auto-loaded by Codex every session. Lean guardrails only
docs/PRD.md (this file)	Implementation authority. Opened on demand
docs/PROMPTS.md	Copy-paste prompts, one per work order, plus drift-recovery
fixtures/abi/ (gitignored)	Real ABI Excel templates and Togonal exports

BUILD_OPS_Blueprint.md — the *why*: evidence, audit, contradictions, reasoning record.

Keep this out of the repo; it contains competitor analysis, an audit of the current build, and a named error in a client-facing document.

What changed in v1.1: every work order is now a self-contained implementation ticket (Context / Do / Do NOT / Files / Acceptance / Rollback), so a session can execute one ticket without reading the whole document. A pre-flight checklist (§0.3) and a drift-recovery protocol (§0.4) were added. No architectural decision changed.

0. How to use this document

This PRD is written to be **seeded into a fresh coding-agent session with no other context**. Everything needed to act is contained here. The reference toolchain is **OpenAI Codex**; the document is tool-agnostic and works with any agent that reads a project instruction file.

Recommended opening prompt:

You are a senior engineer working on BUILD OPS, a construction ERP for Actuate Builders Inc. Read `docs/PRD.md` in full before writing any code. It contains confirmed schema facts, locked architectural decisions, and a numbered work-order sequence. Do not propose rebuilding anything marked LOCKED. Do not introduce entities the PRD explicitly forbids. Start with Work Order WO-01 and confirm your migration plan with me before running it.

The full prompt set lives in `docs/PROMPTS.md`.

Rules for any session working from this PRD:

1. **This is a refactor, not a rewrite.** The existing codebase is sound where it matters (§2). Additive migrations only. No downstream foreign key is re-pointed at any stage.
2. **Do not create a `scope_items` table.** See ADR-03. This is the most likely wrong instinct and it is expensive.
3. Work orders are numbered and ordered. WO-01 → WO-18. Do not skip forward; later orders assume earlier invariants hold.
4. Every work order has explicit **acceptance criteria**. A work order is not done until its criteria pass as automated tests.
5. Where this PRD says **[BLOCKED: O-n]**, the work order cannot be completed without an answer from ABI. Build to the labelled default, mark it configurable, and flag it.
6. All money is `BIGINT` centavos. All percentages are integer basis points. **Never introduce a float into a monetary path.**

0.1 Two ways to run this

Codex CLI (recommended for WO-01 → WO-18). Commit this PRD as `docs/PRD.md` and commit `AGENTS.md` at the repo root. Codex reads `AGENTS.md` automatically at session start and builds its instruction chain once per session; this PRD is opened on demand when a work order touches schema, pricing or migrations.

Start each session with the kickoff prompt from `docs/PROMPTS.md`, then paste the single matching work-order prompt. **One work order per session** — context stays clean and the agent cannot wander into WO-12 while doing WO-04.

A chat assistant (for design review, migration review, and the D-1/D-2 conversations with Commercial). Attach this PRD plus the relevant screenshots. Good for reasoning and for arguing a decision through; do not use it to generate large migrations you then paste in blind.

Four Codex specifics that matter for this refactor:

1. **Do not put this PRD into `AGENTS.md`** . Codex stops adding project guidance once the concatenated instruction files reach `project_doc_max_bytes` (32 KiB by default), and files discovered later are silently dropped. This PRD alone would exceed that. Keep `AGENTS.md` lean and reference `docs/PRD.md` per work order.
2. **`AGENTS.md` is instruction, not enforcement.** It reduces the probability of a destructive migration; it does not remove the possibility. For `DROP` statements and production pushes, use Codex's sandbox permissions, approval settings or `execpolicy` rules. Given WO-04 and WO-05 are structural migrations against real data, configure those before the first migration runs.
3. **Editing `AGENTS.md` mid-session does nothing.** The instruction chain is built once at session start. Restart the session after any edit.
4. **Launch from the repo root.** Codex walks from the git root toward the working directory; starting in a subdirectory can leave root-level instructions outside the walk path.

Verify the instruction file is actually loading before the first work order. Start a session and ask: *"Summarise the hard constraints you're operating under, in five bullets."* If the answer does not mention *no `scope_items`* , *additive migrations only* , and *no general ledger* , the file is not in context — check the size cap and the launch directory.

0.2 The one-paragraph version, for anyone joining mid-refactor

The database spine is already correct: money is integer centavos, tenancy is enforced, and `bom_line_items.id` is a stable UUID that cost, budget, PO and RFQ records already point at. What is wrong is **grain and pricing**. `bom_line_items` mixes work items with material lines at the same level, room names are baked into description strings, and every line is priced at a flat 30% markup instead of a real cost build-up. The refactor adds a `kind` discriminator, a location dimension, and DUPA child tables underneath the existing rows — additive migrations, no foreign key ever moves.

0.3 Pre-flight checklist — do these before WO-01

- **Database backup taken**, restore tested. Every work order below has a rollback, but a backup is the real one.
- **A staging environment exists** with a copy of production data. Backfills (WO-04, WO-05) run on staging first, always.
- **CI runs the test suite on every PR**. If it does not, this is WO-00 — the invariants in §6 are worthless as prose and valuable only as failing builds.
- **The five blocking open questions are sent to ABI** (§11): O-01 VAT base, O-03

Delegation matrix, O-04 Excel templates, O-05 Togal export, O-14 rate owner. They gate WO-06, WO-09 and WO-15. Send them on day one; they take days to come back.

- **One estimator is identified** as the design partner for WO-06/07. The DUPA engine cannot be validated without one.
- **Decide the tenancy/branding question** (O-01b). It is cheap now.

0.4 Drift-recovery protocol

Long refactors drift. If a session starts proposing a rewrite, inventing `scope_items`, or reaching for a general ledger, stop and paste the **Drift Reset** prompt from the Prompt Pack. If output already contains drifted code, discard it rather than repairing it — repaired drift is harder to review than fresh work.

Three red flags that mean stop immediately:

1. Any migration containing `DROP COLUMN`, `DROP TABLE`, or an `ALTER` that re-points an existing foreign key.
2. Any new table named `scope_items`, or any new column named `scope_item_id`.
3. Any monetary value typed as `float`, `double precision`, `numeric` without explicit scale, or a JS `number` used for currency.

1. Context brief (for a cold session)

The company. Actuate Builders Inc. (ABI) is a Philippine design-and-build, fit-out and MEP contractor. Business units: Sales (Business Development), Commercial (Estimation), Design, Procurement/SCM, Service Delivery (SD: Project Manager, Project Engineer, Site Support, Safety Officer, QA/QC), Finance (GA / AR / Treasury), CX, and Management/President.

The product. BUILD OPS is ABI's construction operating system — one connected system from lead to handover. It currently exists as a working build ("Third Code ERP", Next.js on Vercel, Supabase/Postgres) at roughly 25 modules.

ABI's four stages and the documents that gate them:

1. **Proposal** (Sales-led) — PPRF → KYC/financial standing → SI Report + RFI → layout & perspective → RFQ → **BOM + BOE** → Level 1 Master Schedule → **signed BOM** → Contract → NTP + down payment + AR/Project code + CARI
2. **Pre-Construction** (Commercial + SD) — Project Tracker → Planning Phase 1 & 2 → FCD/CSD → Building Admin vetting (15 d) → **LGU building permit (45–60 d)** → CARI →

Allowable Budget → RFPO → PO (BAFO, price comparison, SSJ) → subcontracts → CX onboarding → mobilization

3. **Construction** (SD-led) — kickoffs → materials & goods receipt → WAR with **Thursday cut-off** → weekly meetings → **Variation Orders** (Site Instruction → priced proposal → client sign-off) → QA/QC hold points, IWR, punchlist → **Milestone COC → progress billing**
4. **Handover** — practical handover, COC at 90%/100%, punchlist close-out (30 d), Occupancy Permit, full payment, **construction bond refund**, P&L close-out (30 d), warranty + CX ticketing (24h ack / 48h scheduling) + CNPS

Philippine realities the system must respect: LGU permitting and Building Admin vetting, BIR 2303 and VAT zero-rated certification, CARL and construction-bond lifecycle through to refund, an existing **SAP** goods-receipt touchpoint, the **ABI Delegation of Approval** matrix, and business-day SLAs on every process step.

The core problem. A quantity is measured once in Togonal, then re-typed into a BOE, whose unit rate is re-typed into a BOQ, whose total is re-typed into a proposal, then an Allowable Budget, then a Project Tracker, then an RFPO, then a PO, then a billing forecast, then an invoice — across separately versioned Excel files (`_PPRF_Ver3` , `_SI_Ver1` , `_BOE_Ver2`) owned by five business units. **BUILD OPS's job is to type it once.**

2. Confirmed schema facts (verified against the live database — do not re-litigate)

#	Fact	Status	Consequence
1	Money stored as BIGINT centavos ; percentages as integer basis points . No monetary floats anywhere.	 VERIFIED	The DUPA percentage cascade can be made exactly reproducible. No money-type migration needed.
2	tenant_id is NOT NULL on all application tables ; <code>tenants</code> is the root; RLS scoping already in place.	 VERIFIED	Multi-tenancy is solved. Every new table inherits the same pattern — no exceptions.
3	bom_line_items.id is a stable UUID , already referenced by cost, budget, PO and		The relational spine exists. This is the single most expensive thing to

	RFQ records. Normal saves update/delete by id.	VERIFIED	build and it is already built.
4	<code>scope_item_id</code> does not exist . The system uses <code>bom_line_item_id</code> throughout.	 VERIFIED	See ADR-03. <code>bom_line_items</code> is the Scope Item. Do not rename, do not duplicate.

Therefore: refactor, not rebuild. The three conditions that would have forced a greenfield rewrite — float money, missing tenancy, unstable line identity — all came back clean.

3. Architectural Decision Records (LOCKED)

ADR-01 — Refactor the existing build. LOCKED. Keep: auth, app shell, navigation, tenancy/RLS, Purchase Orders (approval timeline + PH tax handling are production-quality), Inventory (perpetual, immutable movements, UOM, warehouses), Invoices, Claims, KYC Queue, Documents, design system, dashboard metric selection. Everything else is additive or a rewiring of a view over new tables.

ADR-02 — `bom_line_items` is promoted to the Scope Item spine. LOCKED. It already carries a stable UUID referenced by four downstream domains. It stays the anchor. Domain language in the UI may say “Scope Item”; the table does not change name in v1.

ADR-03 — Do NOT create a `scope_items` table. LOCKED. Creating a parallel entity would require re-pointing cost, budget, PO and RFQ foreign keys — a four-domain migration purchased for a naming preference. **Any proposal to introduce `scope_item_id` is rejected by default.**

ADR-04 — The real defect is grain, not naming. LOCKED. `bom_line_items` currently mixes two different levels of abstraction in one table:

- Entrance Hall — Suspended Ceiling, Mineral Fibre 600×600 · sqm · 1 · ₱850 → a **work item**, priced as though it were a material
- Entrance Hall — LED Panel Light 600×600 · pc · 3 · ₱1,500 → a **material line**

A work item is the unit of scope, schedule, progress, billing and margin. A material line belongs *underneath* it, inside the DUPA. The fix is a `kind` discriminator plus DUPA child tables — additive, no FK movement (§5).

ADR-05 — The DUPA is the atomic commercial object. LOCKED. Every downstream number in ABI’s business descends from a unit rate: proposal price, contract value,

allowable budget, procurement target, VO price, margin. The current build prices at a flat 1.30 markup (`COST ₱49,450` → `TCV ₱64,285` , with `MARKUP %` defaulting to 30 in the Add Line form). That is not a cost model. See §4.

ADR-06 — Room/location must leave the description string. LOCKED. Descriptions are currently concatenated: `Entrance Hall — Vinyl Plank Flooring` . The same material appears as three unrelated rows across three rooms, each independently priced and each `unassigned` . Location becomes a dimension; the item becomes a reference. Without this, “total vinyl plank across the project” is unanswerable and procurement cannot consolidate.

ADR-07 — No general ledger. LOCKED. ABI runs SAP. BUILD OPS owns *project* financial truth (commitments, costs, billings, collections forecast, project P&L). SAP remains the statutory book of record. The existing Finance module (ledger accounts, journals, posting periods, cash accounts, reconciliation) is **frozen — no further investment** — pending O-02. Build the GR interface, not a second GL.

ADR-08 — Total is one input to a generic pipeline, never a dependency. LOCKED. The takeoff importer accepts CSV/XLSX with a saved per-source column-mapping profile. A Total export is one such source. Manual entry is always available. The cost chain must never be blocked by one vendor’s export fidelity.

ADR-09 — Money and percentage conventions. LOCKED. `BIGINT` centavos; integer basis points ($8\% = 800$, $7\% = 700$, $12\% = 1200$). **Intermediate DUPA values are computed unrounded and rounded only at presentation and at the persisted unit rate** — this is required to reproduce ABI’s own workbooks (§4.3).

4. The DUPA engine — full specification

This is the heart of the refactor. Build it correctly before anything reads from it.

4.1 Structure

One DUPA per **work item**. Recovered verbatim from ABI’s MNHPI Reefer Facility submission:

```
Header:  Project · Item No./Description · Unit of Measurement · Quantity

A  Material    : item · qty · unit · unit_rate · amount           → Subtotal for Mate
B  Labour      : item · no_of_persons · hourly_rate ·             → Subtotal for Manp
                  productivity_rate_per_hour · amount
C  Equipment   : item · no_of_units · hourly_rate ·             → Subtotal for Equi
                  productivity_rate_per_hour · amount
```

D DIRECT COST = A + B + C
 E INDIRECT COST = OCM/Contingency (8% of D) + Profit (7% of D)
 F VAT = 12% × [base – see 4.3]
 G TOTAL COST = D + E + F
 H UNIT RATE = G ÷ header quantity

Labour and equipment amounts: $\text{amount} = \text{count} \times \text{hourly_rate} \div \text{productivity_rate_per_hour}$. ABI's standard crew is **Foreman / Skilled / Non-Skilled**; observed rates ₱272.02 / ₱199.09 / ₱173.79 per hour. These live in a **crew library**, not typed per sheet.

4.2 Canonical worked example — use as the regression test

From the MNHPI submission, item “Stair footing concrete inc. forms”, unit `cu.m`, header quantity `0.10`:

Section	Detail	Amount
A Material	Stair footing concrete inc. forms — 1.00 cu.m @ ₱5,951.00	₱5,951.00
B Labour	Foreman 1.00 @ ₱272.02, productivity 1.00 → ₱272.02 · Skilled 1.00 @ ₱199.09 → ₱199.09 · Non-Skilled 2.00 @ ₱173.79 → ₱347.57	₱818.685
C Equipment	One-Bagger Concrete Mixer — 1 unit @ ₱600.00/hr, productivity 0.10 → 600 ÷ 0.10	₱6,000.00
D Direct	A + B + C	₱12,769.685 <i>(displays 12,769.69)</i>
E Indirect	OCM 8% of D = ₱1,021.5748 <i>(displays 1,021.57)</i> · Profit 7% of D = ₱893.8780 <i>(displays 893.88)</i>	₱1,915.45
F VAT	12% × D = ₱1,532.3622 <i>(displays 1,532.36)</i>	₱1,532.36
G Total	D + E + F	₱16,217.50
H Unit rate	G ÷ 0.10	₱162,175.00

Test assertion: given these inputs, the engine must return $G = 1621750$ centavos and $H = 16217500$ centavos, exactly.

4.3 Two discrepancies found in ABI's own workbook — both must be handled explicitly

D-1 — The VAT base is labelled $(D+E)$ but computed on D . The sheet's row F reads "12% $(D+E)$ ". Arithmetic proves otherwise: 12% of $(D+E)$ would be ₱1,762.22, but the sheet shows ₱1,532.36, which is exactly 12% of D — and only that value makes G foot to ₱16,217.50. → **Implementation:** `vat_base` is a configurable enum (`direct_only` | `direct_plus_indirect`), defaulting to `direct_only` to reproduce ABI's existing books. Surface the label honestly in the UI. **[BLOCKED: O-01]** — Commercial must confirm which is intended; the difference is ~1.4% of every unit rate.

D-2 — A 10× transcription error is visible in the shipped BOQ. The DUPA yields $H =$ ₱162,175.00 per cu.m. The BOQ line reads `qty 0.10 · unit cost ₱16,217.50 · amount ₱1,621.75`. The estimator carried **G** into the BOQ's unit-cost column instead of **H**, understating that line by a factor of ten — roughly ₱14,600 on one item, in a document that went to a client. → **This is the single most persuasive argument for the whole refactor.** When the BOQ unit cost is *derived* from H rather than typed, this class of error becomes structurally impossible. Ship D-2 as a named regression test.

4.4 Rounding policy

- Compute A, B, C, D, E, F in **unrounded rational arithmetic** (integer centavos with a scale factor, or exact decimal — never float).
- Round **half-up to centavos** only when: persisting `unit_rate_centavos`, persisting `total_centavos`, or rendering.
- H is persisted rounded; all BOQ amounts derive from the persisted H so the client-facing document always foots.
- **Test:** re-deriving a BOQ from persisted rates must reproduce the printed total to the centavo.

4.5 Libraries (what makes the engine fast rather than merely correct)

Library	Contents	Why
Assembly	A reusable DUPA template per work type (e.g. "suspended ceiling, mineral fibre 600×600, sqm") with its material/labour/equipment skeleton and productivity factors	ABI hand-builds ~55 DUPA sheets per project, most structurally identical. This removes 60–80% of estimator keystrokes
		Rates stop being retyped 55

Crew	Foreman / Skilled / Non-Skilled (extensible) with current hourly rates, effective-dated	times and start being maintained once
Equipment	Item, hourly rate, typical productivity	Same
Material catalog	Item, base UOM, current rate, source (RFQ / PO / manual), <code>last_updated_at</code>	Feeds DUPA material lines and gives the staleness flag
Price history	Every quoted and awarded price, by vendor, by date	Makes the rate suggestion defensible rather than a guess

Every rate displays **“last updated”** beside it. A rate older than a configurable threshold (default 90 days) renders with a staleness warning.

5. Schema changes

Migration principles. Additive only. No existing column is dropped in v1. No downstream foreign key is re-pointed. Every new table carries `tenant_id NOT NULL` with the same RLS policy shape as existing tables. Every new table carries `created_at`, `updated_at`, `created_by`, and participates in the audit trail.

M-01 · Discriminate the grain on `bom_line_items`

```
ALTER TABLE bom_line_items
  ADD COLUMN kind text NOT NULL DEFAULT 'work_item'
    CHECK (kind IN ('work_item','material_line')),
  ADD COLUMN parent_line_item_id uuid REFERENCES bom_line_items(id),
  ADD COLUMN location_id uuid REFERENCES project_locations(id),
  ADD COLUMN division_id uuid REFERENCES boq_divisions(id),
  ADD COLUMN item_no text,                -- BOQ numbering, e.g. 'B.2.0'
  ADD COLUMN drawing_revision_id uuid REFERENCES drawing_revisions(id),
  ADD COLUMN takeoff_import_id uuid REFERENCES takeoff_imports(id),
  ADD COLUMN unit_rate_source text NOT NULL DEFAULT 'manual'
    CHECK (unit_rate_source IN ('dupa','manual','client_boq'));

CREATE INDEX ON bom_line_items (tenant_id, parent_line_item_id);
CREATE INDEX ON bom_line_items (tenant_id, kind);
```

Backfill. Classify existing rows by UOM: area/volume/lot units (`sqm`, `cu.m`, `lot`, `lm`, `m2`)
→ `work_item`; discrete/weight units (`pc`, `pcs`, `kg`, `liters`, `set`) → `material_line`. Flag every ambiguous row for estimator review rather than guessing. **No row is auto-**

reparented — attaching a material line to the wrong work item silently corrupts a cost hierarchy. Reparenting is an explicit, reviewed action in the UI (WO-04).

Invariant: `kind = 'material_line' ⇒ parent_line_item_id IS NOT NULL` (enforced by trigger after backfill review completes, not before).

M-02 · Location dimension (ADR-06)

```
CREATE TABLE project_locations (  
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id uuid NOT NULL REFERENCES tenants(id),  
  project_id uuid NOT NULL REFERENCES projects(id),  
  parent_id uuid REFERENCES project_locations(id),    -- building > floor > zone >  
  name text NOT NULL,  
  level text CHECK (level IN ('building','floor','zone','room','area')),  
  sort_order int NOT NULL DEFAULT 0  
);
```

Migration parses the leading "`<Room> -` " prefix out of existing descriptions into `project_locations`, leaving the item description clean. Original string retained in `description_original` for audit.

M-03 · The DUPA tables

```
CREATE TABLE dupas (  
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id uuid NOT NULL REFERENCES tenants(id),  
  bom_line_item_id uuid NOT NULL REFERENCES bom_line_items(id) ON DELETE CASCADE,  
  assembly_id uuid REFERENCES assemblies(id),  
  header_quantity numeric(18,4) NOT NULL,          -- quantity the analysis is built  
  uom text NOT NULL,  
  ocm_bps int NOT NULL DEFAULT 800,                 -- 8%  
  profit_bps int NOT NULL DEFAULT 700,              -- 7%  
  vat_bps int NOT NULL DEFAULT 1200,                -- 12%  
  vat_base text NOT NULL DEFAULT 'direct_only'  
  CHECK (vat_base IN ('direct_only','direct_plus_indirect')), -- D-1  
  direct_cost_centavos bigint NOT NULL DEFAULT 0,  
  indirect_cost_centavos bigint NOT NULL DEFAULT 0,  
  vat_centavos bigint NOT NULL DEFAULT 0,  
  total_cost_centavos bigint NOT NULL DEFAULT 0,  
  unit_rate_centavos bigint NOT NULL DEFAULT 0,     -- H, the value the BOQ consume  
  computed_at timestamptz,  
  UNIQUE (bom_line_item_id)  
);
```

```

CREATE TABLE dupa_material_lines (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id uuid NOT NULL REFERENCES tenants(id),
  dupa_id uuid NOT NULL REFERENCES dupas(id) ON DELETE CASCADE,
  catalog_item_id uuid REFERENCES material_catalog(id),
  description text NOT NULL,
  quantity numeric(18,4) NOT NULL,
  uom text NOT NULL,
  unit_rate_centavos bigint NOT NULL,
  rate_source text CHECK (rate_source IN ('catalog','rfq','history','manual')),
  rate_as_of date,
  sort_order int NOT NULL DEFAULT 0
);

CREATE TABLE dupa_labour_lines (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id uuid NOT NULL REFERENCES tenants(id),
  dupa_id uuid NOT NULL REFERENCES dupas(id) ON DELETE CASCADE,
  crew_role_id uuid REFERENCES crew_roles(id),
  description text NOT NULL,          -- 'FOREMAN','SKILLED','NON-SKIL
  no_of_persons numeric(10,2) NOT NULL,
  hourly_rate_centavos bigint NOT NULL,
  productivity_per_hour numeric(18,4) NOT NULL CHECK (productivity_per_hour > 0),
  sort_order int NOT NULL DEFAULT 0
);

CREATE TABLE dupa_equipment_lines (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id uuid NOT NULL REFERENCES tenants(id),
  dupa_id uuid NOT NULL REFERENCES dupas(id) ON DELETE CASCADE,
  equipment_id uuid REFERENCES equipment_catalog(id),
  description text NOT NULL,
  no_of_units numeric(10,2) NOT NULL,
  hourly_rate_centavos bigint NOT NULL,
  productivity_per_hour numeric(18,4) NOT NULL CHECK (productivity_per_hour > 0),
  sort_order int NOT NULL DEFAULT 0
);

```

Migration of existing material lines. Rows classified `material_line` in M-01 that have a reviewed parent move *down* into `dupa_material_lines` under that parent's DUPA. The original `bom_line_items` row is retained but marked `superseded_by_dupa_line_id` — never hard-deleted, because downstream POs and RFQs may already reference it. Those references remain valid and resolvable.

M-04 • Libraries

`assemblies` (reusable DUPA templates with child template tables mirroring M-03),
`crew_roles` (name, hourly_rate_centavos, effective_from, effective_to),
`equipment_catalog`, `material_catalog` (item, base UOM, current rate, source,
`last_updated_at`), `price_history` (catalog_item_id, vendor_id, quoted/awarded rate,
`source_document`, `occurred_at`), `boq_divisions` (code, name, sort_order, is_preliminaries).

M-05 • Budget and cost baseline

`cost_codes` (derived from BOQ divisions), `budgets` (project_id, status, baseline_locked_at,
`locked_by`), `budget_lines` (budget_id, cost_code_id, **bom_line_item_id**,
`budgeted_centavos`). Commitments and actuals both resolve to `budget_lines`.
bom_line_item_id is the join — not a new key.

M-06 • Process and SLA engine

`process_steps` (code e.g. PR-L, stage, name, responsible_bu, input, input_from, output,
`output_by`, `sla_days`, `is_business_days`, `template_link`, `predecessor_code`),
`task_instances`, `sla_clocks` (`started_at`, `due_at`, `at_risk_at`, `breached_at`,
`paused_reason`), `approval_rules` (object_type, amount_band_low, amount_band_high,
`approver_role`, `sequence`, `escalation_after_days`), `approvals`.

Seeded from ABI's SD Framework deck — roughly 70 steps already specify responsible BU,
input, output and SLA in business days. **That deck is a state machine written in
PowerPoint. Load it as data.**

M-07 • Business-day service

A shared `business_days` service with a Philippine holiday calendar. Every SLA in the deck is
expressed in business days except CX's 24h/48h ticket clocks, which are calendar hours.
Model both.

6. Invariants (enforce as constraints, triggers or tests — not conventions)

#	Invariant	Enforcement
I-01	No float in any monetary path	Lint rule + type check in CI

I-02	Every application table has <code>tenant_id</code> NOT NULL and an RLS policy	Migration test that scans <code>information_schema</code>
I-03	<code>kind='material_line' ⇒ parent_line_item_id</code> IS NOT NULL	Trigger (after backfill review)
I-04	A <code>work_item</code> with <code>unit_rate_source='dupa'</code> has exactly one DUPA	Unique constraint + test
I-05	<code>dupa.unit_rate_centavos</code> recomputes on any child-line change	Trigger or explicit service call; <code>computed_at</code> must be ≥ latest child <code>updated_at</code>
I-06	BOQ line amount = <code>unit_rate_centavos × quantity</code> , always derived, never typed	No editable unit-cost field where <code>unit_rate_source='dupa'</code>
I-07	<code>productivity_per_hour > 0</code>	Check constraint (prevents divide-by-zero)
I-08	A locked budget baseline cannot be edited — only a VO or a logged transfer changes it	Trigger on <code>budgets.baseline_locked_at</code> IS NOT NULL
I-09	PO commitment cannot be created without a <code>budget_line_id</code>	FK NOT NULL on commitment
I-10	Takeoff re-import never deletes and reinserts <code>bom_line_items</code> — it upserts by <code>(takeoff_import_id, source_row_key)</code>	Importer test: assign a vendor, re-run import, vendor survives
I-11	Every state change writes an immutable audit event	Trigger on all mutable tables
I-12	Rounding is half-up to centavos, applied only at persistence and presentation	Unit tests including §4.2
I-13	A BOM cannot be submitted for client approval while its Unresolved queue is non-empty	Guard on the submit action + test
I-14	An AI-drafted line never carries a unit rate until a DUPA is attached	Check constraint on <code>kind='work_item'</code> AND <code>ai_drafted</code>

7. Work orders

Ordered. Each is a self-contained ticket — a session can execute one without reading the rest. Do not skip forward; later orders assume earlier invariants hold.

Global constraints applying to every work order below:

- Additive migrations only. No `DROP`. No foreign key re-pointed.
 - Every new table: `tenant_id NOT NULL`, matching RLS policy, `created_at / updated_at / created_by`, audit participation.
 - No float in any monetary path.
 - Every acceptance criterion ships as an automated test, not a manual check.
-

WO-00 · CI gate (*prerequisite — half a day*)

Context. The invariants in §6 are only real if a build fails when they break. **Do.** Ensure CI runs the full test suite on every PR. Add three static checks: (a) no `float / double precision` in any migration touching money; (b) no new table without `tenant_id NOT NULL`; (c) no record with an `E2E_` prefix in a non-demo tenant. **Acceptance.** A deliberately-broken PR for each of the three checks fails CI.

WO-01 · Purge and separate demo data (*1 day — do this first*)

Context. The live build shows `E2E_QA_20260513_dd8a07a1_acct1` across Accounts, Projects and Documents; four identical `PO-0002` rows in Deliveries; a duplicated `TH/RD CODE FINAL PHASE` lead; and `E2E_QA_..._vendor1` sitting in the production vendor picker beside Daikin Phils. Trading and Powermatic Industries. **Do.** Inventory every QA/E2E-seeded record. Move demo data to a dedicated demo tenant. Purge from the ABI tenant. Fix the duplicate `PO-0002` rows (determine whether they are genuine duplicates or a join fanout in the Deliveries query — check this before deleting anything). **Do NOT.** Delete anything referenced by a real record without first checking the reference. **Files.** Seed scripts, Deliveries query/repository, tenant configuration. **Acceptance.** No `E2E_`-named record is reachable from any ABI-tenant screen. The vendor picker returns only real vendors. Deliveries shows the correct row count for its POs. **Rollback.** Restore from the pre-purge snapshot. *Rationale: management will be shown this build. Every number they see has to be one they believe.*

WO-02 · Business-day service + audit events (2–3 days)

Context. Every SLA in ABI's process deck is in business days; CX ticket clocks are calendar hours. Both are needed before WO-03. **Do.** Build `businessDays.add(date, n) / .between(a, b)` with a Philippine holiday calendar (national regular + special non-working, maintainable as data, not code). Add immutable audit events on all mutable tables: actor, table, row, before, after, timestamp. **Acceptance.** Holiday arithmetic is correct across a year boundary and across Holy Week. Every mutation on a covered table produces exactly one audit row. Audit rows cannot be updated or deleted.

WO-03 · Process & SLA engine, seeded from the SD Framework deck (1–1.5 weeks)

Context. ABI's deck already specifies ~70 steps with responsible BU, input, input-from, output, output-by and SLA in business days. It is a state machine written in PowerPoint. **Do.** Implement M-06 tables. Seed `process_steps` from the deck. Implement clocks with at-risk (80%), breach (100%) and escalation (150%) thresholds. Separate **internal** clocks (enforceable) from **external** clocks (LGU, Building Admin, client — tracked, never escalated against a BU). **Do NOT.** Seed any step whose owner is recorded as a question mark — C-05's "Commercial or SD — PM?" — until the workshop resolves it. Never automate an unowned step. **Acceptance.** A step can be created, assigned, clocked, breached and escalated end-to-end. A process-health view shows breaches by BU. External-clock breaches never generate a BU escalation. **Launch note.** Ship in **observe mode** for 4–6 weeks: report at BU level, not individual level, and tune thresholds with BU heads before enforcement (R-07).

WO-04 · Grain discrimination + reparenting review UI (1 week)

Context. ADR-04. `bom_line_items` currently holds `Entrance Hall — Suspended Ceiling · sqm · 1 · ₱850` (a work item priced as a material) beside `Entrance Hall — LED Panel Light · pc · 3 · ₱1,500` (a material line) at the same level. **Do.** Run migration M-01. Backfill `kind` by UOM: `sqm, cu.m, m2, lm, lot → work_item; pc, pcs, kg, set, liters → material_line`. Route every ambiguous row to a review queue. Build the UI for an estimator to confirm classification and explicitly attach a material line to a work item. **Do NOT.** Auto-reparent anything. Attaching a material line to the wrong work item silently corrupts a cost hierarchy and is very hard to detect later. Do not enable the I-03 trigger until the review queue is empty. **Files.** Migration, `bom_line_items` repository, BOM Builder line list. **Acceptance.** Every existing row is classified or queued. No row was auto-reparented. No downstream FK changed — verify by asserting PO, RFQ, cost and budget row counts and

target ids are identical before and after. **Rollback.** `kind` and `parent_line_item_id` are new nullable columns; reverting is dropping them.

WO-05 · Location dimension (3–4 days)

Context. ADR-06. The same LED panel appears as three unrelated rows across Entrance Hall, GM's Office and Reception Room, each independently priced at ₱1,500 and each `unassigned`. Procurement cannot consolidate; rollup is impossible. **Do.** Run M-02. Parse the leading "`<Room> –` " prefix out of descriptions into `project_locations`. Retain the original string in `description_original`. Build a location picker on the line editor. **Do NOT.** Discard the original description. Parsing is heuristic and will need review. **Acceptance.** Entrance Hall – Vinyl Plank Flooring becomes item Vinyl Plank Flooring at location Entrance Hall. A query returns total vinyl plank sqm across all locations in a project. Unparseable descriptions are queued, not silently mangled.

WO-06 · DUPA engine + libraries (2–3 weeks — the critical path)

Context. ADR-05. This is the heart of the refactor. Currently the build prices everything at flat markup: the Add Line form defaults `MARKUP %` to 30, and `COST ₱49,450` → `TCV ₱64,285` is exactly $\times 1.30$. **Do.** Run M-03 and M-04. Implement §4 exactly: material + labour (crew $\times$ hourly rate $\div$ productivity) + equipment → Direct → OCM 800 bps + Profit 700 bps → VAT 1200 bps → Total → **Unit Rate**. Implement §4.4 rounding: unrounded intermediates, half-up to centavos only at persistence and presentation. Build the assembly, crew, equipment, material-catalog and price-history libraries. Implement D-1's configurable `vat_base`, defaulting to `direct_only`. **Do NOT.** Write any UI before the engine passes the §4.2 test. Do not introduce a float anywhere in the cascade. **Acceptance.**

- §4.2 worked example returns $G = 1621750$ and $H = 16217500$ centavos, exactly.
 - **D-2 regression test:** a BOQ unit cost sourced from G instead of H fails the build.
 - Changing a crew rate in the library recomputes every DUPA referencing it.
 - A DUPA can be built from an assembly template in under 60 seconds.
 - Re-deriving a full BOQ from persisted rates reproduces the printed total to the centavo.
- Blocked by.** O-01 (VAT base) for the default; build configurable and proceed. O-14 (rate owner) for sign-off. *Assign the most senior engineer. Validate against ABI's ~55 MNHPI DUPA sheets before writing UI.*
-

WO-07 · Rebuild BOM Builder as a view over the new model (1.5–2 weeks)

Context. The current screen is a flat table with an editable `UNIT COST` and a `MARKUP %` field. That form is where D-2's 10x error becomes possible. **Do.** New structure: BOQ divisions with sub-totals → work items with **derived** unit rates → DUPA detail behind progressive disclosure. Keep the existing versioning, the "Submit for Client Approval" flow, and the supplier switcher — those instincts are correct. Fix the supplier switcher's rate-card matching to join on `catalog_item_id`, not description text (it currently returns *"No rate cards match this line"* on every AI-drafted row). **Do NOT.** Retain an editable unit-cost field where `unit_rate_source='dupa'`. **Do not keep** `MARKUP %` as a line-level input. **Acceptance.** I-06 holds. The pricing-breakdown chips (RAG / Catalog / Manual / Unpriced) report real state rather than zeros. Margin derives from the DUPA cascade. Rate cards match by foreign key.

WO-08 · Takeoff importer (1.5 weeks)

Context. ADR-08. Total is one producer, never a dependency. **Do.** Build: file → preview → **saved column-mapping profile per source** → validation (UOM recognised? division assigned? duplicate detected?) → **upsert** to `bom_line_items` keyed on `(takeoff_import_id, source_row_key)`. Bind every imported line to a `drawing_revision_id`. **Do NOT.** Delete-and-reinsert on re-import. That orphans every downstream PO and vendor assignment. **Acceptance.** I-10 is the **blocking test**: assign a vendor to an imported line, re-run the same import, the vendor survives. A Total export works as a mapped XLSX on day one. Manual entry remains available.

WO-08a · Disposition of the existing AI CAD auto-draft — **decided, do not re-open**

The current build ships *"Auto-drafted from CAD upload"*, which produces AI-tagged lines such as `Entrance Hall — Suspended Ceiling, Mineral Fibre 600×600 · sqm · 1 · ₱850` with `vendor: unassigned`, and a pricing breakdown reading RAG 0 / Catalog 0 / Manual 0 / Unpriced 0 against 13 lines. The feature **is retained and keeps running**. It is not switched off, and it is not deleted.

What changes: its output stops writing directly to a priced BOM line and instead becomes **one more producer feeding the same pipeline as any other takeoff source**. Concretely:

1. **Route it through the identical validation gate.** AI-drafted rows get no privileged path. Same UOM check, same division assignment, same duplicate detection, same `(takeoff_import_id, source_row_key)` upsert key.
2. **Land every AI row as a work_item in draft state with** `unit_rate_source='manual'`

and no unit rate, rather than as a priced line. An AI-extracted description is a *scope candidate*, not a price. Pricing happens when a DUPA is attached (WO-06).

3. **Unmapped and unpriced rows are visibly queued, never silently accepted.** The current failure mode is a line sitting at qty 1 with an unassigned vendor and no rate card match, looking finished. Every such row appears in an **Unresolved queue** with the reason (no UOM match , no division , no catalog match , quantity defaulted) and a count badge on the BOM header.
4. **Retain the AI provenance tag on the row**, and record the source model, the drawing revision and the extraction timestamp. This is the corpus the post-MVP mapping layer trains and is evaluated against — do not discard it.
5. **A BOM cannot be submitted for client approval while the Unresolved queue is non-empty.** Hard gate. This is the structural fix for lines reaching a client at qty 1 with a defaulted quantity.

Do NOT. Do not build the AI mapping layer in this work order — free-text description normalisation, UOM inference, automatic division assignment and assembly matching are **post-MVP**, first item after the MVP line. Do not let AI-drafted lines bypass validation on the argument that the model was confident. Do not delete the existing feature to “clean up” — its output is the evaluation set.

Acceptance.

- A CAD upload produces `work_item` rows in `draft` , with no unit rate and no vendor, each carrying an `AI` provenance record.
- Every row that fails any validation check appears in the Unresolved queue with a stated reason; none is silently accepted.
- Submitting for client approval is blocked while that queue is non-empty, with the blocking rows listed.
- Re-running a CAD extraction over the same drawing revision upserts — it does not duplicate rows and does not clear vendor assignments (I-10 applies here too).
- A row that has been given a DUPA in WO-06 is not overwritten by a subsequent re-extraction.

Rationale. The AI draft is genuinely valuable at the top of the chain — it turns a drawing into candidate scope in seconds. What made it untrustworthy was that its output *looked* like a finished priced line when it was an unvalidated guess. Keeping the feature while forcing its output through the same gate as every other source preserves the speed and removes the false confidence.

WO-09 · Excel importers (2 weeks — ship alongside WO-06/07, not after)

Context. ABI's operating history is in `_PPRF_Ver3`, `_SI_Ver1`, `_BOE_Ver2` files. These are simultaneously the adoption barrier and the seed corpus — importing historical BOEs populates the assembly, crew and material libraries that make WO-06 fast on day one. **Do.** Importers for PPRF, SI Report, BOQ/BOM, BOE, Level 1 schedule. Each with preview, mapping, validation and a rejected-rows report. **Acceptance.** A real `*_BOE_Ver2_*.xlsx` imports into DUPAs with rates and productivity intact. A real `*_PPRF_Ver3_*.xlsx` creates a client and an opportunity. Nothing imports silently — every skipped row is reported. **Blocked by.** O-04 (the actual templates).

WO-10 · RFQ intake → price comparison → price history → DUPA rates (1 week)

Do. Capture supplier quotes as structured lines. Build the comparison matrix. Write awarded and quoted prices to `price_history`. Surface suggestions inside the DUPA material line with source and date. **Acceptance.** A captured quote updates the catalog rate and appears as a dated, sourced suggestion in the DUPA. Rates older than 90 days render with a staleness warning.

WO-11 · PPRF form + KYC dual-track gate (1 week)

Do. Structured PPRF creating Client + Opportunity. Two independent 2-day tracks: Financial Evaluation Report (Finance-GA; FC recommends → President approves) and Credit Investigation Report (Finance-AR; FC notes → President endorses). Keep the existing KYC Queue UI — the gating concept is already right. **Acceptance.** An opportunity cannot advance past Site Survey until both tracks clear. A flag or rejection locks downstream stages with a visible reason.

WO-12 · Mobile SI Report + RFI (1 week)

Do. Pre-filled from the PPRF. Photo capture. RFI raised from the same screen. Offline-tolerant with sync. **Acceptance.** Completable one-handed on a phone. Survives loss of signal mid-form. Findings attach permanently to the opportunity.

WO-13 · Award automation — the demo moment (1 week)

Context. This is where the value becomes obvious to management. It is also ABI's single most expensive hand-off today. **Do.** Signed BOM → in **one atomic transaction**: create Project, Budget baseline from the approved BOQ, cost codes from BOQ divisions, AR/Project code request, DP invoice draft, CARI task, Project Tracker, CX onboarding task. **Acceptance.** No field is retyped between the signed BOM and the live project. The transaction is atomic and reversible. Demo it to management before proceeding.

WO-14 · Allowable Budget approval + baseline lock (4–5 days)

Do. Approval flow on the Allowable Budget. On approval: lock the baseline, snapshot the original margin. **Acceptance.** I-08 holds — a locked baseline cannot be edited by any path except a VO or a logged transfer with a reason and an actor.

WO-15 · RFPO → PO → budget commitment (1.5 weeks)

Context. The PO module is the best-built part of the current system — the approval timeline (Draft → PM approved → Commercial approved → SCM issued → Supplier notified) and PH tax handling (12% VAT, 2% withholding) are production-quality. Carry them forward unchanged. **Do.** Add the RFPO stage, Delegation-of-Approval routing by amount band, and commitment against `budget_lines` joined on `bom_line_item_id`. **Do NOT.** Rewrite the existing PO approval timeline or tax logic. **Acceptance.** I-09 holds. Issuing a PO reduces remaining allowable on the cost code. A PO that would exceed allowable warns before submission. **Blocked by.** O-03 (the Delegation matrix).

WO-16 · Permits, CARI, bonds + mobilization readiness gate (1.5 weeks)

Context. The LGU permit at 45–60 days is the schedule's critical path and is currently chased over email. Mobilization requires four external returns. **Do.** Permit types: LGU building permit, Building Admin vetting (15 d), DOLE, Occupancy Permit, CARI, and performance/surety/construction bonds through to refund. Each with responsible person, expected return, countdown, escalation. Model LGU duration as min/expected/max **per LGU**, learned from actuals (C-01). **Acceptance.** A single tile shows the four mobilization inputs — commented FCD, PO copies, CARI, NTP from Building Admin — as /  with days-at-risk. Mobilization cannot be marked started until all four land, or an authorised override is logged with a reason and an actor.

WO-17 · Cost control v1 (1.5 weeks)

Do. Budget → committed (POs) → actual (PO-invoiced) → remaining → variance → live margin. Drilldown by cost code. All joins on `bom_line_item_id`. **Acceptance.** Commercial confirms the live margin on a running project matches their own spreadsheet. This is a human sign-off, not a unit test.

WO-18 · Management dashboard v1 (1 week)

Do. Keep the existing metric selection — pipeline TCV, weighted pipeline, GP, blended margin are the right metrics at the right altitude. Add: margin by project with variance, cost variance, permit exposure, unsigned-VO exposure, SLA breaches by BU. **Acceptance.** The President can run a Monday meeting from it without opening Excel.

— MVP line —

Post-MVP, in order: **AI takeoff mapping layer** (description normalisation, UOM inference, division assignment, assembly matching — evaluated against the AI provenance corpus retained in WO-08a) · tender/bid mode (client-issued TOR + BOQ, deviation list, evaluation criteria, subcontractor profile) · schedules L1–L4 with MS Project import + labour reconciliation · progress capture → WAR → Thursday cut-off · milestone → COC → claim → invoice · **VO end-to-end** · QA/QC hold points → IWR → punchlist with plan pinning · GR + inventory issuance → actual cost + SAP interface · subcontracts + vendor performance · handover, O&M, occupancy, bond refund, P&L close-out · warranty + CNPS + CX onboarding · role dashboards · AI layer (DUPA auto-draft → price suggestion → anomaly detection → TOR summarisation → NL search).

8. Automations to implement (MVP subset)

Full set of 44 in the Blueprint §10. These are the MVP-critical ones.

ID	Trigger	Actions
A-01	PPRF submitted	Create Client + Opportunity; open Financial Evaluation (Finance-GA, 2 d) and Credit Investigation (Finance-AR, 2 d); notify FC
A-02	Both KYC reports approved by President	Unlock stage advance past Site Survey; stamp KYC-cleared; log approver
A-	KYC flagged/rejected	Lock downstream stages; alert BD + Management

03		with reason
A-05	Final layout approved	Auto-issue RFQ pack by trade (5 d clock); notify Procurement
A-06	Quote received	Update price comparison + price history; alert Estimator if it beats the DUPA rate by >X%
A-07	Signed BOM uploaded	The award transaction — WO-13
A-11	NTP received	Start Pre-Con clocks: Project Tracker 2 d, Planning Phase 1 ≤ 2 d, Planning Phase 2 18 d, CX onboarding 7 d
A-12	FCD issued for signature	3–5 d client clock → Building Admin vetting 15 d → LGU 45–60 d; notify SD-PE at each return
A-13	Permit / CARI / bond / accreditation nearing expected return or expiry	Alert responsible person + BU head; escalate on breach
A-14	Allowable Budget approved	Lock baseline; snapshot original margin; enable procurement
A-15	RFPO signed	Generate PO drafts from BOM lines; route per Delegation; start 18 d clock
A-16	PO issued	Commit to cost code; reduce remaining allowable; create expected Delivery; warn if commitment exceeds allowable
A-17	All four mobilization inputs received	Mark Mobilization Ready; notify PM; unlock Construction stage
A-30	Any cost, commitment or VO change	Recompute forecast-at-completion and live margin; alert if margin drops below threshold
A-41	Any SLA at 80% / 100% / 150%	Notify assignee → BU head → Management; add to SLA-breach dashboard
A-42	Approval required	Route per Delegation by amount band and object type; auto-escalate on timeout
A-44	Any state change	Immutable audit event

9. Risks and guardrails for the refactor

#	Risk	Guardrail
R-01	Estimators keep using Excel	WO-09 ships alongside WO-06/07. MVP success test: an estimator produces a client-presentable BOQ + BOE <i>faster in BUILD OPS than in Excel</i> on launch day. If false, do not launch — fix the engine and the import
R-02	DUPA engine under-built	Most senior engineer; validated against the ~55 MNHPI sheets before any UI work; §4.2 and D-2 as permanent regression tests
R-03	Scope creep back into the GL	ADR-07 is LOCKED. Any GL work requires an explicit written reversal
R-04	Backfill mis-classifies grain	Nothing auto-reparents. Ambiguous rows go to a human review queue (WO-04)
R-05	Stale master data poisons every DUPA	<code>last_updated_at</code> beside every rate; staleness warning at 90 days; a named owner for rate maintenance [BLOCKED: O-14]
R-06	The deck is mid-revision — automating an unowned step	Resolve C-02/C-03/C-05 (§10) in a short workshop before WO-03 seeding. Never automate a step whose owner is recorded as a question mark
R-07	SLA enforcement experienced as surveillance	Launch in observe mode 4–6 weeks; BU-level reporting before individual; thresholds tuned with BU heads before enforcement
R-08	External SLAs outside ABI's control	Separate <i>internal</i> (enforceable) from <i>external</i> (trackable) clocks. Never escalate a BU for an LGU delay
R-09	Importer regenerates rows and orphans downstream records	I-10 is a blocking test on WO-08
		Launch by stage on live projects: Proposal-stage first

R-10	Big-bang launch	(Commercial + Sales only), then Pre-Con, then Construction. One pilot project fully through before company-wide rollout
R-11	Single-estimator dependency in the assembly library	Multi-estimator review; library entries versioned with an owner

10. Contradictions carried forward (flagged, not silently resolved)

#	Contradiction	Default taken	Needs
C-01	LGU permit 45–60 days (SD Framework) vs 30 days (Yamaha schedule)	Model as min/expected/max per LGU , learned from actuals	O-12
C-02	Contract turnaround “30 days after signing of BOM and NTP” vs “2 weeks (max) upon receipt of NTP”	14 business days (the tighter commitment)	O-08
C-03	Level 4 Master Schedule listed as output of both Planning Phase 1 and Phase 2	Phase 1 = endorsement package; Phase 2 = the schedule	Workshop
C-04	Step E annotated “ <i>Originally from Construction Stage. Proposed to move to Pre-construction</i> ”	Build the step stage-configurable	Workshop
C-05	Level 1 Master Schedule owner recorded as “ Commercial or SD – PM? ” — question mark in the source	SD-PM produces, Commercial approves	Workshop
C-06	Two intake modes: ABI-generated BOM vs client-issued BOQ (MNHPI tender)	Support both; <code>unit_rate_source='client_boq'</code> exists from day one	Post-MVP
C-07	Yamaha schedule signed by SPCI staff, no ABI signatory	—	O-01

C-08	SAP goods receipt vs the build's own GL	ADR-07: interface, don't replace	O-02
C-09	Handover at 100% (SD Framework) vs partial per-zone handovers (Yamaha)	Handover is per-zone/package	Confirm
D-01	DUPA row F labelled "12% (D+E)" but computed on D	<code>vat_base='direct_only'</code>	O-01, blocking WO-06
D-02	BOQ unit cost taken from G instead of H — a live 10x error	Unit cost always derived from H	Permanent test

11. Open questions

Blocking specific work orders:

- **O-01** — Is the DUPA VAT base *direct cost only* or *direct + indirect*? Differs by ~1.4% on every unit rate. → **blocks WO-06**
- **O-03** — The actual **ABI Delegation of Approval** matrix: amount bands, approver roles, sequences, escalation. → **blocks WO-15**
- **O-04** — The real Excel templates: `PPRF`, `SI`, `BOE`, `Milestone Definition`, `Project Tracker`, `Allowable Budget Form`, `Interim Payment Certificate`, `Level 1 MS`. (*The SD deck's "Link to Template" column is the checklist.*) → **blocks WO-09**
- **O-05** — A real **Togal export file** — format, columns, fidelity. → blocks Togal AI mapping
- **O-14** — Who owns rate maintenance as a standing responsibility? Without a name, R-05 happens. → blocks WO-06 sign-off

Needed before later phases:

- **O-02** — Which SAP modules does ABI actually run (GR only? AP? GL? full FI/CO)?
- **O-06** — Are 8% OCM / 7% profit ABI policy, DPWH convention, or MNHPI-specific?
- **O-07** — Standing cost-code/WBS chart across projects, or per-project divisions?
- **O-09** — What Level 2 and Level 3 schedules contain, and who owns them
- **O-10** — Retention terms and standard warranty period (MNHPI TOR says minimum 1 year — is that ABI's standard?)

- **O-11** — Concurrent project count and target user count
- **O-12** — Which LGUs does ABI most often work with?
- **O-13** — Does ABI want a vendor/subcontractor portal, or should external parties stay on email?
- **O-01b** — Is BUILD OPS ABI's internal system or a Th/rd Code product with ABI as tenant? (*Tenancy is already built correctly either way — this is a branding and commercial decision.*)

Assumptions in force (labelled, cheap to reverse): A1 contract turnaround 14 business days · A2 Phase 1 = endorsement, Phase 2 = schedule · A3 L1 schedule by SD-PM · A4 SAP remains statutory book of record · A5 Delegation is amount-banded and sequential (prepared → reviewed → noted → approved) · A6 Dupa defaults 800/700/1200 bps, configurable per project/client · A7 crew = Foreman/Skilled/Non-Skilled · A8 cost codes derive from BOQ divisions · A9 milestone percentages project-configurable · A10 handover per-zone · A12 all deck SLAs are business days except CX's 24h/48h · A13 existing PH tax treatment (12% VAT, 2% withholding, retention, output VAT on posting) is correct and carried forward unchanged.

12. Definition of done for the refactor

The refactor is complete when all of the following are true:

1. No `E2E_`-named record is reachable from any ABI-tenant screen.
2. The §4.2 worked example returns `G = 1621750` and `H = 16217500` centavos, exactly, as an automated test.
3. The D-2 regression test — BOQ unit cost derived from `H`, never `G` — is permanent and passing.
4. No `bom_line_items` row mixes grain: every material line has a reviewed parent work item.
5. Room has left the description string; a project-wide material rollup query returns correct totals.
6. A takeoff re-import preserves vendor assignments (I-10).
7. A real historical `_BOE_Ver2_.xlsx` imports with rates and productivity intact.
8. A signed BOM creates a project, budget baseline and cost codes with **zero fields**

retyped.

9. A locked budget baseline cannot be edited except through a VO or a logged transfer.
10. Issuing a PO commits against a budget line and reduces remaining allowable.
11. Live margin on a running project is confirmed correct by Commercial against their own spreadsheet.
12. An ABI estimator produces a complete, client-presentable BOQ + BOE **faster in BUILD OPS than in Excel.**

Item 12 is the only one that matters commercially. The other eleven exist to make it true.

14. Document control

Document	Version	Role
BUILD_OPS_Blueprint.md	1.1	Evidence and reasoning record — the <i>why</i> . Audit of the current build, AS-IS/TO-BE, contradictions, competitor analysis, AI map
docs/PRD.md	1.3	Implementation authority — the <i>what</i> and <i>in what order</i>. Where the two disagree, this document wins
BUILD_OPS_Prompt_Pack.md → docs/PROMPTS.md	1.1	Execution aid — copy-paste prompts, one per work order, plus drift-recovery and review prompts
AGENTS.md	1.0	Codex instruction file, repo root. Auto-loaded each session. Lean by design

Change log.

- *v1.0 (08 Aug 2026)* — initial PRD following the refactor-vs-rebuild decision.
- *v1.1 (12 Aug 2026)* — work orders restructured as self-contained tickets; WO-00 (CI gate) added as a prerequisite; pre-flight checklist and drift-recovery protocol added (§0.3, §0.4); no architectural decision changed.
- *v1.3 (12 Aug 2026)* — retargeted to **OpenAI Codex** as the reference toolchain:
AGENTS.md replaces CLAUDE.md , file paths updated to docs/PRD.md and docs/PROMPTS.md , and Codex-specific operational notes added (§0.1) covering the 32

KiB instruction cap, instruction-vs-enforcement limits, session restart on edit, and launch directory. No architectural decision changed.

- *v1.2 (12 Aug 2026)* — **WO-08a added**, settling the disposition of the existing AI CAD auto-draft: retained and running, but routed through the same validation gate as every other takeoff source, landing as unpriced draft scope with an Unresolved queue and a hard submit gate. Invariants I-13 and I-14 added. Post-MVP AI mapping item clarified. No architectural decision changed.
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13. Final direction

Purge the test data. Load ABI's own process deck as the workflow engine. Build the DUPA engine on top of the `bom_line_items` spine that already exists — fixing the grain, not the name. Ship the Excel importers alongside it, because those files are simultaneously the adoption barrier and the seed corpus. Then demonstrate the award transaction, where a signed BOM becomes a live project with a locked budget and nothing retyped.

Do not build the general ledger. Do not build the knowledge graph. Do not build the AI layer yet. Each is defensible alone, and each competes for exactly the engineering time the DUPA engine needs.

Type it once. Everything else follows.